GREEN ROOF THERMAL SIMULATION

Complete Technical Documentation

CAM (Bromelia) vs C3 (Wedelia)

Finite Volume Method | Chagolla-Aranda et al. (2025)

Teknik Mesin — Universitas Indonesia
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Chapter 1: Overview & Research Novelty

This document covers the complete thermal simulation of a green roof system comparing CAM (Bromelia) and C3 (Wedelia) plants. The model is based on Chagolla-Aranda et al. (2025), adapted and calibrated for tropical-humid conditions at Universitas Indonesia, Jakarta, using local soil–rice husk substrate.

1.1 Experimental Setup

Instrument	Data Available	Period	Key Variables
Davis Vantage Pro 2	✓ Available	3 Mar – 24 Apr 2026	T_a, G_sol, RH, u, rain
NI Temperature Sensor	✓ Available	3 Apr – 10 Apr 2026 (7.1 days)	20 channels, 1 min interval
Licor 6800	✓ C3 only (April 2026)	Several sessions	gsw, A, TleafEB
Licor 6800 CAM	■ Older files (months ago)	Pre-experiment	r_stoma_min CAM placeholder
Soil Moisture	■ Not extracted yet	Phase 2 only	Depth 2cm and 7cm

1.2 The Four Novelty Claims

Novelty 1 (STRONGEST)	First explicit comparison of CAM vs C3 green roof thermal performance in tropical-humid climate (Jakarta) using numerically calibrated FVM model.
Novelty 2	Local substrate: soil + rice husk (sekam padi) instead of imported pumice. Rice husk is abundant agricultural waste in Indonesia.
Novelty 3	Dynamic T_in from measured NI sensor data (24.8–42.8 degC, mean 29.5 degC) instead of the constant 25 degC assumed in the reference paper.
Novelty 4	LAI calibration via inverse problem: dual-target for CAM (T_g_top + T_s_in), single-target for C3 (T_s_in only). Quantifies uncertainty via sensitivity analysis.

Chapter 2: Code Architecture Flowchart

The simulation is structured in 6 sequential phases. Each step is labeled INIT (initialize), DO (execute), CALC (calculate), or OUTPUT (store result).

PHASE 1: Define Parameters (Seg 1–5)	
INIT	PlantParameters: H_f, LAI, rho_f, tau_f, alpha_f, r_stoma_min — for CAM and C3
INIT	SubstrateParameters: lambda_dry, rho_g, cp_g, theta_sat, k_theta_sat, b, psi_sat
INIT	SlabParameters: standard concrete values, H_slab
INIT	GeometryParameters: H_g, H_slab, T_in_default=29.5C (not 25C!) [C1]
INIT	NumericalParameters: dt=1s, Nz_sub=107, Nz_slab=67 [from paper Sec 3.1]

PHASE 2: Load & Prepare Data (Seg 6–7)	
DO	load_weather_data(3-24april.txt) — cols: T_a(2),RH(5),u(7),rain(17),G_sol(19)
DO	load_NI_sensor_data(xlsx) — XML parser [C5], filter T2A anomalies [C6]
DO	synchronize_datasets() — align Davis 1min grid with NI LabVIEW timestamps [C7]
CALC	Thermo helpers: P_sat (Magnus), T_sky (Swinbank), P_a, VPD, gamma=66.8 Pa/K

PHASE 3: Main Simulation Loop — 604,800 timesteps x 1 second (Seg 8–12)	
READ	Weather at step t: T_a_K, G_sol, RH, u, rain -> j_pr
GET	[C1] T_in_current from NI T1Ka(CAM) or T3Kd(C3) — not 25C constant
CALC	BLOK 1 Foliage: r_a(A.2-A.4), r_stoma(A.6-A.10), h_conv_f(A.1), h_eva_f(A.5)
CALC	R_f_net(Eq.2,3), solve T_f_new implicit [Eq.1]
CALC	BLOK 2a Substrate heat: (rho_cp)_g,lambda_g,k_theta(B.8-B.11)
CALC	q_top BC [Eq.5,6] -> FVM 107 nodes -> TDMA -> T_g[0..106] [Eq.4,7]
CALC	BLOK 2b Substrate moisture: D_theta(B.11), j_eva(Eq.10)
CALC	Richards PDE [Eq.8] + BC [Eq.9,11] -> TDMA -> theta[0..106]
CALC	BLOK 3 Slab: pure conduction [Eq.12], BC slab-substrate [Eq.13]
OUTPUT	BC bottom h_in*(T_s-T_in) [Eq.14 C1] -> TDMA -> T_s[0..66] -> q_s_in
STORE	Save every 60 steps: T_f,T_g_top,T_g_bot,T_s_in,theta,q_s_in,j_eva,T_in_used

PHASE 4: Run Both Plants (Seg 16)	
RUN	run_simulation(weather, bromelia, ...) -> results_CAM
RUN	run_simulation(weather, wedelia, ...) -> results_C3
CALC	Q_gain = trapz(q_s_in, dx=60) [Eq.18] — negative = heat sink

PHASE 5: Calibration & Analysis (Seg 13–14)	
INIT	LAI_bounds=(0.1, 5.0), T_in_series from NI data [C1]

DO	Brent method (scipy): ~10-20 iterations to find optimal LAI
CALC	CAM dual-target: MSE(T_g_top vs T1Ke) + MSE(T_s_in vs T1Tb)
CALC	C3 single-target: MSE(T_s_in vs T3Ka)
DO	Sensitivity: LAI=[0.3,0.5,0.8,1.0,1.5,2.0,3.0] -> T_s_in_max, Q_gain, ET

PHASE 6: Output & Visualization (Seg 15–16)

PLOT	(a) Temp CAM | (b) Temp C3 | (c) Heat flux | (d) VWC
PLOT	(e) Evapotranspiration | (f) [C1] T_in actual vs 25C assumption
OUTPUT	Comparison table: Q_gain, T_s_in_max, theta_final, ET_mean

Chapter 3: Segment-by-Segment Code Explanation

Segment 1 — PlantParameters

Parameter / Function	Description
name / plant_type	Label and "CAM"|"C3". plant_type controls stomata logic in compute_stomatal_resistance().
H_f	[REAL] Canopy height. Bromelia=0.098m, Wedelia=0.276m. Enters Eq.A.2-A.4 (aerodynamic resistance).
d_f	[ESTIMATE] Leaf width in meters. Bromelia≈0.045m, Wedelia≈0.060m. Enters drag coefficient B.6.
LAI	[REAL] Leaf Area Index. Bromelia=1.95, Wedelia=1.07. Most sensitive parameter — calibrated.
rho_f [NEW]	[REAL] Leaf reflectivity. Bromelia=0.390, Wedelia=0.438. Used to compute alpha_f.
alpha_f [C3]	[CORRECTED] = 1 - rho_f - tau_f. Bromelia=0.540, Wedelia=0.362 (was 0.462 before tau_f update).
epsilon_f	=0.95. Standard for all green leaves. Constant, no measurement needed.
tau_f [C2]	[CORRECTED] C3=0.20 (not 0.10!), CAM=0.07. C3 thin leaves transmit more solar radiation.
r_stoma_min	[C4] C3=167.2 s/m from Licor REAL data. [C8] CAM=400.0 PLACEHOLDER — needs Licor CAM file.

Segment 2 — SubstrateParameters

Parameter / Function	Description
lambda_dry	[LIT] 0.08 W/mK. Dry thermal conductivity. Rice husk is insulator — lower than pumice.
rho_g	[TODO-LAB] Bulk density. Expect 800–1100 kg/m3 for soil+rice husk. Specific gravity test.
cp_g	[LIT] 1300 J/kgK. Organic soil mix. Enters B.8: rho_cp=(0.2+theta)*rho_g*cp_g.
theta_sat	[TODO-LAB] Saturation VWC = porosity. Expect 0.50–0.65 for rice husk mix. Water content test.
k_theta_sat	[TODO-LAB] MOST CRITICAL. Saturated hydraulic conductivity. Expect 3e-6 to 4e-5 m/s. Permeability test.
psi_sat	[LIT] 0.35m. Air-entry potential. Consistent with loam from Rawls et al. (1982).
b	[LIT] 5.5. Brooks-Corey exponent. k_theta=k_sat*(theta/theta_sat)^(2b+3). b=5.5 consistent with loam.
theta_min	=0.05 m3/m3. Wilting point default. OK for most substrates.

Parameter / Function	Description
l_fg	=2.45e6 J/kg. Latent heat of vaporization. Every 1g water evaporated carries 2450J away.

Segment 3 — SlabParameters

Parameter / Function	Description
lambda_s	=1.74 W/mK. Standard concrete. Constant (unlike substrate).
rho_s / cp_s	=2300 kg/m3 / 840 J/kgK. Standard concrete. rho_cp_s=1,932,000 J/m3K.
alpha_s / epsilon_s	=0.40 / 0.82. For Reference Roof model only — not used in GR simulation.
H_slab	[TODO-MEASURE] Slab thickness. Only parameter needing direct measurement.

Segment 4 — GeometryParameters [C1]

Parameter / Function	Description
H_g	[TODO-MEASURE] Substrate depth. Insert rod until resistance felt. Measure 3-5 points, average.
T_in_default [C1]	[CORRECTED] =29.5+273.15K. Mean from NI data T1Ka. NOT 25C as in paper. Impact on q_s_in: ~3x!
h_in	=8.0 W/m2K. Standard convection coefficient for horizontal surface. No change needed.

Segment 5 — NumericalParameters

Parameter / Function	Description
dt	=1.0s. Paper Section 3.1: time-step independent result verified.
Nz_substrate	=107. dz=H_g/106≈0.75mm per node. Mesh-independent verified in paper.
Nz_slab	=67. dz=H_slab/66≈1.5mm per node.
convergence	=1e-5. Gauss-Seidel criterion. Stop when change < 0.00001 between iterations.

Segment 6 — Data Loaders [C5][C6][C7]

Parameter / Function	Description
load_weather_data()	Read TXT Davis: sep=tab, skip 2 headers. Cols: T_a(2),RH(5),u(7),rain(17),G_sol(19).
load_NI_sensor_data() [C5]	XML parser via zipfile — NOT extract-text (truncates at 1000 rows). Reads all 9032 rows.
T2A filter [C6]	66 anomalous values (< -10C or > 80C) -> NaN -> linear interpolation. Max cluster check.

Parameter / Function	Description
LabVIEW timestamp [C7]	epoch = 1904-01-01. Offset = days since Jan 1 1904. Convert to datetime.
synchronize_datasets()	Find overlap period, resample NI to 1min aligned grid with Davis.
extract_r_stoma_min() [C4]	r_stoma=1/(gsw*0.0224). Filter A>0 and gsw>0.001. Take 5th percentile. C3=167.2 s/m.

Segment 7 — Thermodynamic Helpers

Parameter / Function	Description
saturation_pressure(T_K)	Magnus equation: P_sat=610.78*exp(17.269*T_C/(T_C+237.29)). Exponential with temperature.
sky_temperature(T_a,RH)	epsilon_sky=0.711+0.56*RH+0.73*RH^2, T_sky=epsilon^0.25*T_a. Jakarta high RH -> T_sky≈T_a.
ambient_vapor_pressure()	P_a=RH/100*P_sat(T_a). VPD=P_f_sat-P_a. Low VPD in Jakarta limits ET.
psychrometric_constant()	gamma=cp_air*P_atm/(0.622*l_fg)≈66.8 Pa/K. Converts pressure to temperature driving force.

Segment 8 — Block 1: Foliage Model

Parameter / Function	Description
compute_aerodynamic_resistance()	Eq.A.2-A.4: d0=0.701*H_f^0.975, Z0=0.131*H_f^0.997, r_a=[ln(ratio)]^2/(k^2*u).
compute_stomatal_resistance()	Eq.A.6: r_stoma=(r_stoma_min/LAI)*f1*f2*f3*f4. CAM: x12 daytime, x0.5 nighttime [C8].
f1-f4 stress functions	Eq.A.7-A.10: f1=light, f2=temperature, f3=VPD, f4=soil moisture. All >=1 (penalize).
h_conv_f vs h_eva_f	Eq.A.1,A.5: conv=LAI*(rho*cp)/r_a. eva=LAI/gamma*(rho*cp)/(r_a+r_stoma). Stomata affect only h_eva.
solve_foliage_temperature()	Eq.1,2,3: energy balance implicit solve. Linearize P_sat -> direct algebra, no iteration.

Segment 9 — Block 2: Substrate Model

Parameter / Function	Description
compute_substrate_properties()	All 4 properties depend on theta every timestep: rho_cp_g(B.8), k_theta(B.10), lambda_g(B.9), D_theta(B.11).
k_theta nonlinearity	k=k_sat*(theta/theta_sat)^14 for b=5.5. At theta=0.5*theta_sat: k drops to 0.00006*k_sat!
solve_substrate_heat()	Eq.4,5,6,7: FVM 107 nodes, lambda at interface (not node), q_top BC, continuity slab BC.

Parameter / Function	Description
solve_substrate_moisture( )	Eq.8,9,10,11: Richards PDE. j_eva=j_eva_foliage+j_eva_substrate. CAM siang: j_eva_foliage~=0.

Segment 10 — Block 3: Slab Model [C1]

Parameter / Function	Description
solve_slab_heat()	Eq.12,13,14: pure conduction, lambda_s constant. BC top=continuity, BC bottom=h_in*(T_s-T_in).
T_in_K parameter [C1]	New optional parameter. If provided: use dynamic T_in from NI. If None: use T_in_default=29.5C.
q_s_in = KEY OUTPUT	q_s_in = h_in*(T_s[-1]-T_in). Positive=heat INTO interior (bad). Negative=heat OUT (good!).

Segment 11 — TDMA Solver

Parameter / Function	Description
Thomas Algorithm	O(n) vs O(n^3) for matrix inversion. Called 3x per timestep: T_g(107), theta(107), T_s(67).
Forward sweep	Eliminate lower diagonal a[i] from top to bottom. Result: upper bidiagonal.
Back substitution	From x[-1]=d[-1], work backwards: x[i]=d[i]-c[i]*x[i+1].
Safeguard	denom=max(denom, 1e-30). Prevent division by zero when substrate very dry.

Segment 12 — Main Simulation Loop [C1]

Parameter / Function	Description
T_in_series parameter [C1]	New: pass NI timeseries for T_in. Resampled to 1s. Fallback to T_in_default if None.
T_in_used stored	New: save T_in actually used each minute. For comparison plot vs 25C paper assumption.
Q_gain post-processing	Eq.18: trapz(q_s_in_array, dx=60). Negative=net heat sink. Paper GR: -227.8 J/m2.

Segment 13 — LAI Calibration

Parameter / Function	Description
calibrate_LAI()	Brent method: minimize objective(LAI) using scipy.optimize.minimize_scalar, xatol=0.01.
CAM dual-target	MSE(T_g_top sim vs T1Ke NI) + MSE(T_s_in sim vs T1Tb NI). More constrained -> better result.

Parameter / Function	Description
C3 single-target	MSE(T_s_in sim vs T3Ka NI) only. Less constrained -> wider uncertainty range.

Segment 14 — Sensitivity Analysis

Parameter / Function	Description
sensitivity_analysis_LAI()	Run with LAI=[0.3,0.5,0.8,1.0,1.5,2.0,3.0]. Record T_s_in_max, Q_gain, ET_mean.
Why needed	LAI uncertainty after calibration -> quantify how much output changes. Supports thesis defensibility.

Segment 15 — Visualization [C1]

Parameter / Function	Description
6-panel layout	(a) Temp CAM + NI overlay, (b) Temp C3 + NI overlay, (c) heat flux, (d) VWC, (e) ET, (f) T_in.
Panel (f) [C1]	NEW: T_in_actual (CAM T1Ka, C3 T3Kd) vs 25C paper assumption. Shows why correction matters.

Segment 16 — Main Block

Parameter / Function	Description
Usage	Load data -> sync -> fill params -> calibrate LAI -> plot. Full pipeline in sequence.
Status printout	Displays all [C1]-[C8] corrections applied and TODO parameters at runtime.

Chapter 4: All Formulas — Chronological Execution Order

Every formula called during one simulation timestep (dt=1s), in execution order. Paper equation numbers are from Chagolla-Aranda et al. (2025).

Code ID	Formula / Expression	Paper Eq.	Role in Model
F0.1	dz_sub = H_g / 106 dz_slab = H_slab / 66	Sec 3.1	Grid spacing for FVM nodes
F1.1	T_a_K = T_a_C + 273.15	—	Kelvin conversion for radiation
F1.2	j_rain = rain_mm_per_min / 60	—	Rain -> mass flux [kg/m2s]
F2.1	P_sat = 610.78 * exp(17.269*T_C / (T_C+237.29))	—	Magnus: saturation vapor pressure [Pa]
F2.2	P_a = (RH/100) * P_sat(T_a)	—	Actual ambient vapor pressure [Pa]
F2.3	VPD = P_f_sat - P_a	—	Driving force for evapotranspiration
F2.4	eps_sky = 0.711+0.56*(RH/100)+0.73*(RH/100)^2 T_sky = eps_sky^0.25 * T_a	—	Effective sky temperature [K]
F2.5	gamma = 1005*101325 / (0.622*2.45e6) = 66.8 Pa/K	—	Psychrometric constant
F3.1	d0 = 0.701 * H_f^0.975	Eq. A.3	Displacement height [m]
F3.2	Z0 = 0.131 * H_f^0.997	Eq. A.4	Roughness length [m]
F3.3	r_a = [ln((z_ref-d0)/Z0)]^2 / (k^2*u)	Eq. A.2	Aerodynamic resistance [s/m]
F3.4	f1 = 1 + exp(-0.034*(G_sol-3.5))	Eq. A.7	Stomata: light response function
F3.5	f2 = (exp(0.3*T_C)+258) / (exp(0.3*T_C)+27)	Eq. A.8	Stomata: temperature response
F3.6	f3 = 4e-3 + exp(-0.73 * 0.622e3/101325 * VPD)	Eq. A.9	Stomata: VPD response
F3.7	f4 = theta_sat / theta	Eq. A.10	Stomata: soil moisture stress
F3.8	r_stoma = (r_stoma_min/LAI)*f1*f2*f3*f4 CAM siang: *12.0, malam: *0.5 [C8]	Eq. A.6	Actual stomatal resistance [s/m]
F3.9	h_conv_f = LAI*(rho_air*cp_air)/r_a	Eq. A.1	Foliage convection coefficient [W/m2K]
F3.10	h_eva_f = (LAI/gamma)*(rho_air*cp_air)/(r_a+r_stoma)	Eq. A.5	Foliage evaporation coefficient [W/m2Pa]
F3.11	eps_fg = 1/(1/eps_f + 1/eps_g - 1)	Eq. (3)	Effective foliage-substrate emissivity
F3.12	R_f_net = alpha_f*G_sol - eps_f*sigma*(T_f^4-T_sky^4) - eps_fg*sigma*(T_f^4-T_g^4)	Eq. (2)	Net radiation on foliage [W/m2]
F3.13	T_f_new = B/A [implicit solve, Eq.1 linearized]	Eq. (1)	Foliage temperature [K]
F4.1	(rho_cp)_g = cp_g*(0.2+theta)*rho_g	Eq. B.8	Volumetric heat capacity substrate [J/m3K]
F4.2	k_theta = k_sat*(theta/theta_sat)^(2b+3)	Eq. B.10	Hydraulic conductivity [m/s]
F4.3	lambda_g = lambda_dry + theta*k_theta	Eq. B.9	Thermal conductivity substrate [W/mK]
F4.4	u_f = u * ln((H_f-d0)/Z_M)/ln((Z_u-d0)/Z_M)	Eq. B.5	Wind velocity above canopy [m/s]
F4.5	a_drag = (0.28*LAI*H_f*d_f)^0.5	Eq. B.6	Canopy drag coefficient

Code ID	Formula / Expression	Paper Eq.	Role in Model
F4.6	u_g = u_f * exp(-a_drag*(1-0.05/H_f))	Eq. B.4	Wind under canopy [m/s]
F4.7	r_c = 1/(0.004 + 0.012*u_g)	Eq. B.3	In-canopy resistance [s/m]
F4.9	h_conv_g = LAI*(rho_cp)_air/(r_a+r_c) h_eva_g = (LAI/gamma)*(rho_cp)_air/(r_a+r_vap)	Eq. B.1,B.2	Substrate heat/evap coefficients
F4.10	R_g_net = (1-rho_g)*tau_f*G_sol + eps_fg*sigma*(T_f^4-T_g^4)	Eq. (6)	Net radiation on substrate
F4.11	q_top = R_g_net - h_conv_g*(T_g-T_a) - h_eva_g*(P_g_sat-P_a)	Eq. (5)	Surface flux at z=0
F4.12	FVM: a_w[i]=lam_w/dz^2, a_p[i]=(rho_cp)/dt+a_w+a_e, d[i]=(rho_cp)/dt*T_g[i]	Eq. (4)	Substrate heat FVM discretization
F4.13-15	TDMA: forward sweep + back substitution -> T_g[0..106]	Eq. (7)	Substrate temp profile [K]
F5.1	D_theta = -b*k_sat*psi_sat/theta * (theta/theta_sat)^(b+3)	Eq. B.11	Hydraulic diffusivity [m2/s]
F5.4	j_eva = j_eva_f + j_eva_g = h_eva*(P_sat-P_a) [f,g each]	Eq. (10)	Total evapotranspiration [kg/m2s]
F5.5	Richards PDE: dtheta/dt = d/dz(D_theta*dtheta/dz) + dk_theta/dz	Eq. (8)	Moisture transport PDE
F5.6	BC top: (D_theta*dtheta/dz)|_0 = j_pr - j_eva	Eq. (9)	Moisture BC at surface
F5.7	BC bot: drainage if theta>=theta_sat, else no-flow	Eq. (11)	Moisture BC at drain
F6.1	(rho_cp)_s = rho_s*cp_s = 1,932,000 J/m3K (constant) FVM: a_w=a_e=lambda_s/dz^2	Eq. (12)	Slab conduction — constant lambda
F6.2	BC top: lambda_g[-1]*T_g[-1]/dz^2 source term	Eq. (13)	Continuity flux at slab-substrate
F6.3	BC bot: a_p[-1] += h_in/dz, d[-1] += h_in*T_in_current/dz [C1]	Eq. (14)	Convection to interior — dynamic T_in
F6.4	q_s_in = h_in * (T_s[-1] - T_in) [KEY OUTPUT]	Eq. (14)	Heat flux to interior [W/m2]
F8.1	Q_gain = trapz(q_s_in_array, dx=60) [J/m2]	Eq. (18)	Net heat gain over simulation period

Chapter 5: Corrections Applied [C1]–[C8]

[C1] T_in DYNAMIC from NI Sensor — CRITICAL

Problem: The paper assumes T_in=25C constant (air-conditioned room). Actual NI data shows T_in=24.8–42.8C (mean 29.5C) because the experimental box is outdoor and unconditioned.
Solution: Changed T_in_default=29.5C. Added T_in_series parameter to run_simulation() and solve_slab_heat(). T_in taken from T1Ka (CAM) or T3Kd (C3) every timestep.
Impact: Impact on q_s_in: ~3x difference (56 W/m2 at 25C vs 20 W/m2 at 29.5C for same T_s).

[C2] tau_f Updated for Both Plants — CRITICAL

Problem: Paper uses tau_f=0.10 for Sedum (CAM succulent). Applied blindly to Wedelia C3.
Solution: Wedelia C3 (thin 0.25-0.45mm leaf): tau_f=0.20. Bromelia CAM (thick 1-3mm + wax): tau_f=0.07. Based on PROSPECT-D model and CAM vs C3 leaf morphology literature.
Impact: Substrate under Wedelia receives ~3x more solar radiation from canopy transmission.

[C3] alpha_f Recalculated from rho_f — CRITICAL

Problem: Previous alpha_f was assumed. rho_f was measured directly from the plants.
Solution: alpha_f = 1 - rho_f - tau_f. Bromelia: 1-0.390-0.07=0.540. Wedelia: 1-0.438-0.20=0.362.
Impact: alpha_f now physically consistent with actual measured leaf reflectivity.

[C4] r_stoma_min C3 from Real Licor Data — CRITICAL

Problem: Previous value was dummy 150.0 s/m.
Solution: Extracted from Licor 6800 data April 7 2026 (10:12 session). Filter: gsw>0.001 and A>0. Take 5th percentile of r_stoma = 1/(gsw*0.0224). Result: 167.2 s/m.
Impact: C3 real stomatal resistance now represents actual Wedelia physiology.

[C5] NI Loader: XML Parser Instead of extract-text

extract-text truncates output at 1000 rows. NI data has 9032 rows (7.1 days).

[C6] T2A Anomalies Filtered (66 rows)

66 values in T2A (Tanah Bawah CAM) have values like -67 billion C. Sensor was disconnected temporarily. Fixed by NaN + linear interpolation.

[C7] Timestamp Synchronization Davis-NI

Davis records at exact minutes (xx:00). NI LabVIEW has drift (e.g. 14:58:16). Fixed by resample(1min).mean() on NI data to align with Davis grid.

[C8] r_stoma_min CAM Still Placeholder

Licor CAM files from several months ago not yet extracted. Current value 400.0 s/m is a conservative estimate. Needs real Licor CAM data.

Chapter 6: Supporting & Opposing Literature (25 Papers)

#	Authors (Year)	Journal/Source	Key Finding	Position	Novelty
P1	**Cao et al. (2019)**	Building & Env.	CAM lowest canopy cooling (ET 2.3 mm/day) vs C3 (4.6 mm/day) vs C4 (5.4 mm/day)	SUPPORT	N1
P2	**Lundholm et al. (2010)**	PLOS ONE	Mixed plant functional groups outperform monocultures for GR ecosystem services	SUPPORT	N1
P3	**D'Arco et al. (ISHS)**	ISHS Proc.	Explicit comparison C3 vs CAM-facultative for GR cooling performance	SUPPORT	N1
P4	**Mansfield & Blanusa (2018)**	Urban Ecosystems	ET rates 3.5x higher in warm/dry vs cool/humid — species differences matter more in humid	SUPPORT	N1,N3
P5	**Stovin et al. (2019)**	Energy & Buildings	Model comparison: RMSE 1.2–3.0C benchmark for GR temperature simulation	SUPPORT	benchmark
P6	**Poe et al. (2021)**	J. Clean. Prod.	Carbonized rice husk improves water holding capacity, bulk density, porosity of GR substrate	SUPPORT	N2
P7	**Convertino et al. (2021)**	Green & Sust. Chem.	Loam+rice husk better than expanded clay for thermal comfort and plant growth	SUPPORT	N2
P8	**Chabannes et al. (2024)**	Discover Materials	Rice husk thermal conductivity: 0.070–0.129 W/mK	SUPPORT	N2
P9	**Dinh et al. (2025)**	Sci. Tech. Built Env.	Rice husk concrete: thermal time lag 2h, damping factor 0.6	SUPPORT	N2
P10	**Budhiyanto & Tampubolon (2025)**	JARDS (Indonesia)	GR in Indonesia: 91% heat flux reduction, 13.14C temperature drop, 60% energy savings	SUPPORT	N3
P11	**Aziz et al. (2023)**	MDPI Sustainability	Indonesia lags behind Singapore/Malaysia in GR adoption — research gap exists	SUPPORT	N3
P12	**Zingre et al. (2018)**	Solar Energy	GR thermal simulation in Singapore (RH 84%, T_mean=28C) — similar to Jakarta	SUPPORT	N3
P13	**Phoemphoonthanyakit (2023)**	ScienceDirect	GR in warm humid climate: 84% heat transfer reduction, 12C lower temperature	SUPPORT	N3
P14	**Chagolla-Aranda (2025)**	J. Build. Eng.	Main reference paper. FVM model, Sedum CAM, pumice substrate, Mexico climate	REFERENCE	base
P15	**Johannessen thesis (2021)**	Carleton Univ.	Inverse model calibration of GR thermal properties. RMSE 0.41–1.0 W/m2	SUPPORT	N4

#	Authors (Year)	Journal/Source	Key Finding	Position	Novelty
P16	**Pfoser et al. (2017)**	MDPI Sustainability	LAI has LARGEST effect on thermal performance (LAI=5 -> 25C reduction vs LAI=0.01)	SUPPORT	N4
P17	**Zhang et al. (2018)**	Energy & Buildings	Variable LAI reduces error from 39.3% to 7.1% vs constant LAI	SUPPORT	N4
P18	**MDPI Buildings (2025)**	MDPI Buildings	Identifies LAI and stomatal resistance as most influential — calibration needed	SUPPORT	N4
P19	**Djedjig et al. (2012)**	Int. Comm. H&M; Transf.	Coupled heat+mass transfer GR model validated. 19-day temporal limit.	SUPPORT	FVM
P20	**Quezada-Garcia (2018)**	ScienceDirect	FVM without linearization validated with experimental data	SUPPORT	FVM
P21	**Wang et al. (2023)**	ScienceDirect	Coupled heat-water FD model, validated 10-day experiment. 10 days sufficient.	SUPPORT	FVM
P22	**Sailor (2008) via review**	Energy & Buildings	Simpler models give similar predictions — why use complex FVM?	QUESTION	method
P23	**Van Mechelen et al.**	ScienceDirect	GR already effective in tropical humid climate — what is new contribution?	QUESTION	N3
P24	**Lazzarin et al.**	ScienceDirect	Sedum CAM switches to C3 when water abundant — CAM/C3 distinction not always clear	QUESTION	N1
P25	**Stout & Rowe (2014)**	ScienceDirect	CAM plants may not contribute to stormwater management compared to C3	QUESTION	scope

Chapter 7: Variable Gap Analysis — Literature vs. Your Values

For parameters not yet measured in laboratory, this chapter provides literature reference values and justifies the placeholder estimates used in the simulation.

Parameter	Your Value	Literature Range	Source Paper	Status	Comparison
lambda_dry [W/mK]	0.08	0.04–0.20 (organic+lightweight)	Sailor & Hagos (2011) Lunt et al. (2022)	LIT	At lower bound — justified because rice husk is ligh
rho_g [kg/m3]	[TODO-LAB]	700–1100 (rice husk mix)	Poe et al. (2021) Sailor & Hagos (2011)	TODO	Expect 800–1000 kg/m3. Rice husk lighter than pur
cp_g [J/kgK]	1300	840–1500 (organic soil)	Sailor & Hagos (2011)	LIT	Valid for organic-rich substrate. Higher than minera
theta_sat [m3/m3]	[TODO-LAB]	0.43–0.65 (GR substrates)	Stovin et al. (2020) Rawls et al. (1982)	TODO	Estimate 0.55. Rice husk adds macropores -> highe
k_theta_sat [m/s]	5e-6 [PLACEHOLDER]	3e-6 – 4e-5 (GR substrates)	Andres-Domenech (2021) Rawls et al. (1982)	LIT	Placeholder at lower bound of range. Loam value=3
psi_sat [m]	0.35	0.22–0.79 (loam range)	Rawls et al. (1982)	LIT	Consistent with Loam (0.478m) and Sandy Loam (0
b (Brooks-Corey)	5.5	4.9–6.3 (loam-sandy loam)	Rawls et al. (1982)	LIT	Very close to Loam (5.39) from Rawls table. Justifie
r_stoma_min CAM [s/m]	400.0 [PLACEHOLDER]	150–2000 (CAM species)	Steinfort et al. (2018) Quezada-Garcia (2023)	C8-TODO	400 s/m = malam hari range bawah CAM. Siang ha
r_stoma_min C3 [s/m]	167.2 [REAL]	50–200 (C3 herbs)	Quezada-Garcia (2023) Licor 6800 data	REAL	Measured from actual Licor data April 7, 2026. Perc
theta_initial [m3/m3]	0.8 * theta_sat	60–80% saturation (standard assumption)	Wang et al. (2023)	LIT	Standard initial condition when soil moisture data u

7.1 Thesis Framing for Literature Parameters

For parameters still relying on literature values, use this framing in the methodology chapter:

```
"Parameter substrat yang belum tersedia dari pengujian laboratorium
 diestimasi menggunakan data literatur untuk campuran tanah-bahan organik.
 Thermal conductivity kering (lambda_dry=0.08 W/mK) diadopsi dari
 Sailor & Hagos (2011) dan Chabannes et al. (2024) untuk substrat organik
 ringan berbasis agricultural waste. Parameter hidraulik (theta_sat, k_sat,
 b, psi_sat) diestimasi dari Rawls et al. (1982) untuk kelas tekstur loam.
 Sensitivitas model terhadap ketidakpastian parameter ini dianalisis melalui
 sensitivity analysis LAI."
```

Chapter 8: Quick Reference Tables

8.1 Paper Equation Numbers

Eq. #	What it describes	Implemented in
(1)	Foliage energy balance ODE	solve_foliage_temperature()
(2)	Net radiation on foliage	solve_foliage_temperature()
(3)	Effective foliage-substrate emissivity	solve_foliage_temperature()
(4)	Substrate heat PDE	solve_substrate_heat() — FVM
(5)	Boundary condition top substrate	solve_substrate_heat() — node 0
(6)	Net radiation on substrate surface	solve_substrate_heat() — R_g_net
(7)	Boundary condition bottom substrate	solve_substrate_heat() — node -1
(8)	Richards equation for moisture	solve_substrate_moisture() — FVM
(9)	Moisture BC top	solve_substrate_moisture() — node 0
(10)	Total evapotranspiration flux	solve_substrate_moisture() — j_eva
(11)	Moisture BC bottom (drainage)	solve_substrate_moisture() — node -1
(12)	Slab conduction PDE	solve_slab_heat() — FVM
(13)	BC top slab (continuity)	solve_slab_heat() — node 0
(14)	BC bottom slab (convection to interior)	solve_slab_heat() — node -1, q_s_in
(18)	Net heat gain integral	run_simulation() — post-loop
A.1–A.5	Foliage heat/evap coefficients	compute_aerodynamic_resistance()
A.6–A.10	Stomatal resistance and stress functions	compute_stomatal_resistance()
B.1–B.7	Substrate convection/evap under canopy	solve_substrate_heat()
B.8–B.11	Substrate moisture-dependent properties	compute_substrate_properties()

8.2 NI Sensor Channel Mapping

Channel	Physical Location	Model Variable	Used For
T1Ke	Tanah Atas CAM	T_g_top_CAM	Calibration target: Bromelia dual (1 of 2)
T1Tb	Atap Indoor CAM	T_s_in_CAM	Calibration target: Bromelia dual (2 of 2)
T3Ka	Atap Indoor C3	T_s_in_C3	Calibration target: Wedelia single
T1Ka	Ruangan CAM	T_in_CAM [C1]	Dynamic T_in for CAM simulation
T3Kd	Ruangan C3	T_in_C3 [C1]	Dynamic T_in for C3 simulation
T2A	Tanah Bawah CAM	T_g_bot_CAM [C6]	66 anomalies -> interpolated
T2Ka	Atap Indoor RR	T_r_in_RR	Reference conventional roof
T1Ta	Atap Outdoor CAM	T_s_ext_CAM	Outdoor surface temperature CAM

8.3 Parameter Status Summary

Parameter	Value	Source	Status
H_f bromelia	0.098 m	Measurement 1yr ago	REAL
H_f wedelia	0.276 m	Measurement 1yr ago	REAL
LAI bromelia	1.95	Measurement 1yr ago -> calibrate	REAL+CALIB
LAI wedelia	1.07	Measurement 1yr ago -> calibrate	REAL+CALIB
rho_f bromelia	0.390	Direct measurement	REAL
rho_f wedelia	0.438	Direct measurement	REAL
tau_f bromelia [C2]	0.07	PROSPECT-D + CAM morphology	LIT
tau_f wedelia [C2]	0.20	PROSPECT-D + C3 morphology	LIT
alpha_f bromelia [C3]	0.540	1-0.390-0.07	CALCULATED
alpha_f wedelia [C3]	0.362	1-0.438-0.20	CALCULATED
r_stoma_min C3 [C4]	167.2 s/m	Licor 6800 April 2026	REAL
r_stoma_min CAM [C8]	400.0 s/m	Literature estimate	PLACEHOLDER
lambda_dry	0.08 W/mK	Rice husk literature	LIT
cp_g	1300 J/kgK	Organic soil literature	LIT
rho_g	TBD	Soil mechanics lab	TODO
theta_sat	TBD	Soil mechanics lab	TODO
k_theta_sat	TBD	Soil mechanics lab (CRITICAL)	TODO
H_g	TBD	Direct measurement at box	TODO
H_slab	TBD	Direct measurement at box	TODO
d_f both plants	TBD	ImageJ from photos	TODO
T_in [C1]	Dynamic from T1Ka/T3Kd	NI sensor data	REAL

8.4 Answers for Common Prof Questions

Q: What is the novelty?

First CAM vs C3 green roof thermal comparison in tropical-humid Jakarta using calibrated FVM model with local substrate (tanah+sekam padi) and dynamic T_in.

Q: Why calibration, not validation?

Limited data: soil moisture unavailable for Phase 1, plants declined in Phase 2, CAM Licor only from months ago. Calibration framing is more scientifically honest.

Q: Why FVM and not simpler model?

FVM allows dual-target LAI calibration against two independent sensor locations (T_g_top AND T_s_in). Enables detailed substrate temperature profile analysis.

Q: Is 7 days of data sufficient?

Wang et al. (2023) validated a similar model with 10-day data. 7 days captures at least 7 full diurnal cycles — sufficient for diurnal pattern characterization.

Q: How do CAM and C3 differ in the model?

Single line: if plant_type=="CAM" and G_sol>50: r_stoma *= 12.0. This reduces daytime ET by ~12x for CAM, which is the core thermal difference between species.

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